

To determine threshold minimum magnetising inductance value for the Flyback converter.

1st Assume 100% efficient power conversion

$$\text{So } P_s = P_o.$$

$$\text{This gives } V_s I_s = \frac{V_o^2}{R} \quad \text{where } I_s = \frac{\overbrace{(\overbrace{I_{Lm}}^{\text{Average } L_m \text{ Current}}) DT}^{\text{Average source current}}}{T}$$

$$\text{So } V_s I_{Lm} D = \frac{V_o^2}{R}$$

$$\Rightarrow I_{Lm} = \frac{V_o^2}{V_s D R} = \frac{V_s D}{(1-D)^2 R} \left(\frac{N_2}{N_1} \right)^2 \quad \text{using } V_o = \frac{V_s D}{(1-D)} \left(\frac{N_2}{N_1} \right)$$

for $I_{Lm \min}$

$$\frac{V_s D}{(1-D)^2 R} \left(\frac{N_2}{N_1} \right)^2 = \frac{1}{2} \cdot \frac{V_s D}{L_m f} \quad \leftarrow \Delta I_{Lm} \text{ closed.}$$

$$\Rightarrow L_m f_{s \min} = \frac{(1-D)^2 R}{2} \left(\frac{N_1}{N_2} \right)^2$$